

A Frequency Dial for Structure-Conditioned Diffusion

Anonymous submission

Abstract

A structure-conditioned diffusion adapter is trained for one operating point. Canny edges, a depth map, a sketch: each fixes how tightly the output must follow the condition, and moving that trade-off means training again. We train a single adapter on conditions whose information content is randomised — a high-pass filter whose cutoff r is drawn per sample — and read r back at test time as a dial. On FFHQ-512, structure consistency runs from 0.7469 at $r=0.05$ to 0.1110 at $r=0.7$ while sample diversity rises from 0.3126 to 0.4334, and FID stays between 52.77 and 57.40 across the whole range. Turning the adapter’s strength down instead reaches the same structure consistency at a cost of up to 104.95 FID against 53.97 for the frequency dial, so the two knobs are not interchangeable.

Two findings came out of building it. The frozen SD VAE destroys a high-pass condition before any adapter can read it: only 0.5773 of the condition survives an encode–decode round trip at $r=0.3$ and 0.0700 at $r=0.7$. And placement dominates capacity — residuals on the decoder skip connections carry 0.5370 of the structure signal against 0.2112 for cross-attention injection, while quadrupling adapter width buys far less. The adapter trains 6,611,469 parameters, $55\times$ fewer than ControlNet.

1 Introduction

ControlNet [34] and its successors [12, 17, 24, 37] solved conditional control and then froze it. An adapter trained on Canny edges follows Canny edges; one trained on depth follows depth. The strength of that following is a property of the trained weights, so an artist who wants the same sketch honoured more loosely has to pick a different checkpoint or accept whatever the one they have was trained to do.

We change what the condition is rather than how hard the model listens to it. The condition is the image after a radial high-pass filter with normalised cutoff r . At $r=0$ the model receives the grayscale image and the task is colourisation. As r grows the filter removes more of the spectrum, until at $r=1$ almost nothing is left: measured on FFHQ, a cutoff of 0.05 already discards 97.5 % of the spectral energy. Training draws r per sample, so one set of weights sees the whole range, and r becomes an inference-time parameter.

Our contributions are:

- A control dial. One adapter spans structure consistency 0.7469 to 0.1110 with FID flat between 52.77 and 57.40 (Sec. 4.1).
- Evidence that the dial is not a rescaled version of adapter strength. At matched structure consistency

the strength knob costs 104.95 FID where the frequency knob costs 53.97 (Sec. 4.2).

- A measurement of how much of a high-pass condition survives the frozen VAE, which explains why latent-space conditioning fails for sparse structure (Sec. 4.3).
- A controlled comparison of injection sites showing that placement matters more than adapter capacity or receptive field (Sec. 4.4).

2 Related work

Conditional control. Text-to-image diffusion [6, 18, 26, 27, 29] became spatially controllable with ControlNet [34], which duplicates the encoder and injects residuals through zero convolutions. T2I-Adapter [17] does the same job with a small feature extractor, IP-Adapter [32] adds decoupled cross-attention for image prompts, and GLIGEN [13] grounds generation on boxes. Uni-ControlNet [37] and UniControl [24] fold many condition types into one model, and ControlNet++ [12] closes a consistency loop on the produced image. Each of these fixes the condition type and, with it, how tightly the output must follow: the operating point is a property of the trained weights. Our cutoff moves that point at inference on one set of weights.

Efficient control on transformer backbones. As backbones moved from UNets to diffusion transformers [1, 20], control moved from duplicated encoders to lightweight modules. ControlNeXt [21] removes the parallel branch, OminiControl [30] reuses the backbone itself as the condition processor, and EasyControl [36] and NanoControl [15] inject conditions through LoRA-style [7] modules at a small fraction of the parameter cost. Recent work also examines where conditions should enter: Laytrol [9] on preserving pretrained knowledge under layout control, PixelControl [14] on condition fidelity, and Zhou et al. [38] on removing redundant attention in multi-condition DiTs. Our injection-site comparison (Sec. 4.4) reaches a compatible conclusion on a UNet backbone: placement dominates capacity.

Frequency in diffusion. Frequency decomposition appears in diffusion mostly as an intervention on the model’s own signals rather than on its condition. FreeU [28] rebalances skip and backbone features spectrally at no training cost, FreSca [8] scales frequency bands of the noise prediction and, like us, uses an energy-based cutoff, DeCo [16] decouples frequencies in pixel-space diffusion, and FeRA [33] routes adaptation by frequency energy. FOCUS [31] conditions the reverse process on learned frequency priors for test-time adaptation. Closest to us in spirit, Prasain [23] condition on wavelet bands instead of

edge maps for map-to-satellite synthesis. What none of these do is randomise the condition’s bandwidth during training so that the band becomes a control at inference, which is the mechanism we study.

Structure preservation. StructDiff [3] targets structure-preserving spatially controllable generation, and sketch-conditioned pipelines have long used edge maps as the structural carrier. Our condition is a continuum rather than a fixed extractor, which is what makes a dial possible.

Training and evaluation. We train with classifier-free guidance [5] dropout and Min-SNR loss weighting [2], and evaluate with FID [4] under the clean-FID protocol [19], LPIPS [35] and CLIP score [25] on FFHQ [10] with BLIP [11] captions. SDXL and SD3 [1, 22] are the natural next backbones.

3 Method

Condition. Given an image x , we convert to grayscale, take the 2D FFT, zero every frequency inside a disc of radius $r \cdot \min(H, W)/2$, invert, and min-max normalise to $[0, 1]$. The normalisation is deliberate: it removes the absolute scale of the filter response, so the model has to infer structure density from spatial statistics. That is what lets a hand drawing, which has no r , be used at test time.

Condition encoder. A strided convolutional stem reads the condition image directly at 512^2 and reduces it $8\times$. We do not encode the condition with the frozen VAE; Sec. 4.3 shows why.

Injection. Encoder features enter the UNet in two places. Zero-initialised 1×1 heads produce one residual per decoder skip tensor and one for the mid block, added through the UNet’s existing ControlNet interface. In parallel, each cross-attention site attends over a 3×3 window of the condition features, gated by a zero-initialised per-channel vector. Both paths start as exact no-ops: with them switched off the network reproduces stock Stable Diffusion 1.5 [26] bit for bit, which we assert before every training run.

Training. SD1.5 [26] stays frozen. We train 6,611,469 parameters on 65,100 FFHQ-512 [10] images with BLIP [11] captions, batch 32, bf16, 12 epochs, on one H200. r is drawn uniformly from $[0, 1]$ per sample.

4 Experiments

We evaluate on 3,500 held-out images, 50 sampler steps, CFG 7.5, seed 2026. We report FID [4] with the clean-FID protocol [19], LPIPS [35] and CLIP score [25]. Structure consistency (SC) re-extracts the condition from the generated image at the same cutoff and correlates it with the condition the model was given. Feeding the ground truth back gives exactly 1.000 at every r ; an unrelated face gives 0.122 at $r=0$ and ≈ -0.02 above it.

r	SC	Diversity	LPIPS	CLIP	FID
0.00	0.6991	0.2994	0.3341	29.54	50.13
0.05	0.7469	0.3126	0.3495	29.41	51.58
0.10	0.6211	0.3414	0.3748	29.53	54.68
0.20	0.4019	0.3719	0.4003	29.47	53.45
0.30	0.2347	0.3794	0.4148	29.73	52.77
0.50	0.1259	0.4100	0.4418	29.78	53.97
0.70	0.1110	0.4334	0.4625	29.92	53.69
1.00	0.0010	0.5066	0.5329	30.30	57.40

Table 1: Operating curve, 3,500-image test split, 500 sampled.

4.1 The dial

Table 1 gives the operating curve. Structure consistency falls monotonically from $r=0.05$ and sample diversity rises monotonically, which is the trade-off the dial is supposed to expose. FID stays between 50.13 and 57.40 across the entire sweep, so the range is usable end to end rather than degrading into noise.

Figure 3 holds the seed and the caption fixed and varies only r . Figure 4 plots the curve and, below it, both dials in the same structure-consistency/FID plane.

Across seeds 2026, 2027 and 2028 the curve is stable: SC at $r=0.05$ is 0.7479 ± 0.0015 and at $r=0.3$ it is 0.2318 ± 0.0106 .

4.2 Strength is not the same knob

An obvious alternative is to train on one cutoff and scale the adapter’s contribution at inference. We tested it on the same checkpoint, holding $r=0.1$ and scaling both injection paths.

Setting	SC	Diversity	FID
scale 0.25	0.0408	0.7752	104.95
scale 0.50	0.2146	0.6061	76.46
scale 0.75	0.4893	0.4171	58.90
$r=0.3$ (ours)	0.2347	0.3794	52.77
$r=0.5$ (ours)	0.1259	0.4100	53.97

Table 2: Strength dial against frequency dial. Reading across matched SC: scale 0.50 reaches 0.2146 at FID 76.46, while $r=0.3$ reaches 0.2347 at FID 52.77.

Weakening the adapter degrades the image; narrowing the condition’s bandwidth does not. The frequency parameterisation is what makes the trade-off traversable.

4.3 The VAE eats the condition

Encoding the condition with the frozen SD VAE, as an IP-Adapter-style [32] design would, loses most of it. Figure 5 plots it. Passing the condition through encode-decode and re-measuring leaves 0.8562 at $r=0.1$, 0.5773 at $r=0.3$ and 0.0700 at $r=0.7$. An adapter reading that latent cannot recover information the autoencoder has already discarded. Building the condition at 256 and upsampling moves the structure to lower absolute frequency and preserves far more (0.7978 at $r=0.3$), which is a data-pipeline artefact

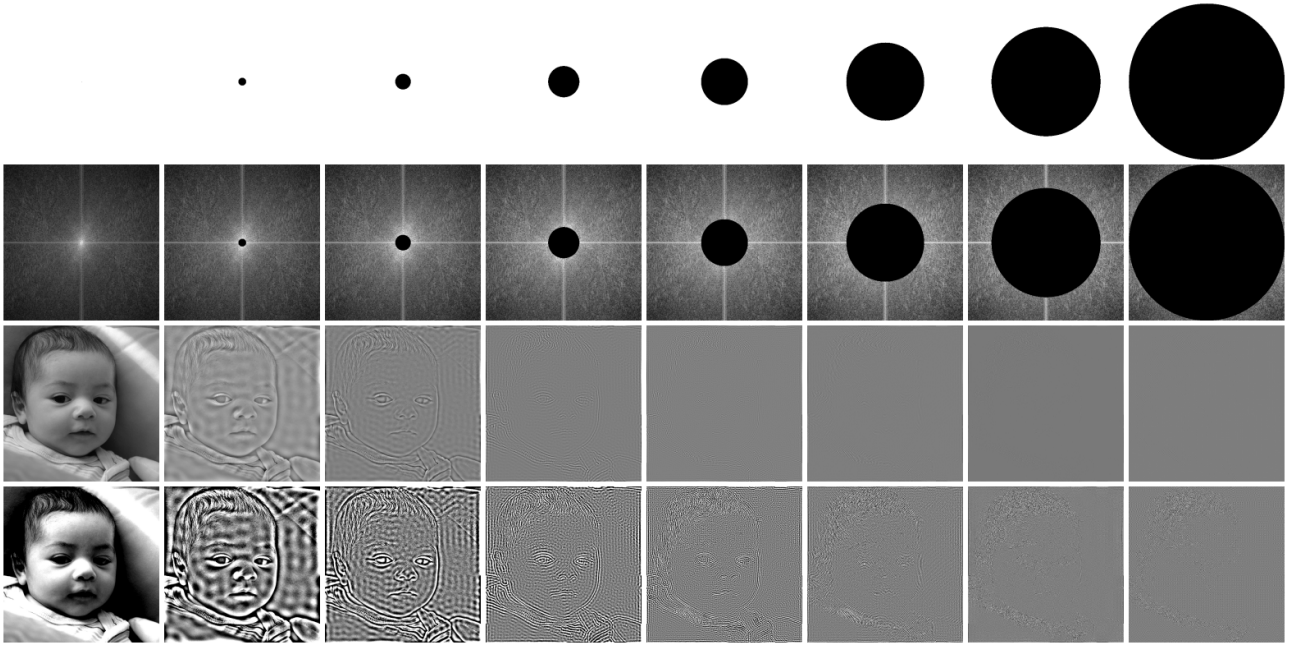


Figure 1: The dial, from the frequency side. Columns are $r = 0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0$. **Row 1:** the radial mask; black is removed. **Row 2:** the log-magnitude spectrum after masking, with the hole growing outward. **Row 3:** the resulting condition, exactly as the model receives it. **Row 4:** the same data displayed at $\pm 2\sigma$ so that the residual content is visible. Above $r \approx 0.2$ the condition looks like flat grey — its standard deviation falls from 0.196 at $r=0$ to 0.041 at $r=1$ — yet the face outline survives in row 4, and the model still reaches SC 0.4019 at $r=0.2$. Low contrast is not the same as no information, which is what the per-image normalisation forces the model to learn.

worth knowing about when comparing against older setups.

4.4 Where to inject

Starting from the same encoder and training schedule, we varied one field at a time (Table 3). Skip-connection residuals dominate. Figure 6 decomposes them. Ablating them away and keeping only cross-attention drops SC at $r=0$ from 0.6991 to 0.2112; keeping only the skip path retains 0.5370. With both paths off the model collapses to 0.0368 SC and FID 120.50, which is the control that says the adapter is doing the work.

Capacity helps much less: widening the projection to a 128-channel stem raises SC at $r=0.05$ from 0.1485 to 0.2826, and doubling again to 256 reaches only 0.2986. A 7×7 attention window is worse than 3×3 (0.0359).

Variant	SC $r=0.05$	SC $r=0.1$	FID $r=0.1$
VAE condition, attn only	0.0234	0.0147	76.25
+ zero-init gate	0.0303	0.0177	65.71
Conv condition encoder	0.1485	0.0772	56.39
+ 128-ch stem	0.2826	0.1882	53.17
+ 256-ch stem	0.2986	0.2158	52.84
+ 7×7 window	0.0359	0.0218	58.51
+ skip residuals	0.7469	0.6211	54.68

Table 3: One field changed per row against the row above it.

4.5 Against per-cutoff specialists

We trained four adapters at fixed r and compared each against the single randomised- r model at its own cut-off. The result is mixed: the specialist wins at $r=0.1$ (0.7049 against 0.6211, $z = +17.5$) and marginally at $r=0.7$ ($z = +2.6$), the single model wins at $r=0.3$ (0.2347 against 0.1214, $z = -19.0$), and $r=0.5$ is a tie ($z = -0.7$).

One model therefore does not dominate four. What it does instead is give a dial the specialists cannot: a model trained at $r=0.1$ has never seen $r=0.3$, so its curve is a single point.

4.6 Baselines

Against pretrained ControlNet [34] on its own condition, over the same test images and captions: ControlNet-Canny reaches Edge-F1 0.6854, LPIPS 0.5294, FID 71.92 with 361,279,120 trainable parameters; ControlNet-HED reaches 0.5733 / 0.6548 / 118.98. At $r=0.1$ our adapter reaches Edge-F1 0.7439, LPIPS 0.3748 and FID 54.68 with 6,611,469 parameters. The conditions differ — Canny edges against a frequency band — so this is a comparison of operating points, not a like-for-like control.

4.7 Hand-drawn sketches

Figure 7 shows the transfer. On 40 real face drawings the model was never trained on, structure consistency against the input drawing is 0.5484 and CLIP score is 23.43. FID is not reported: 40 images is too few for a stable estimate, and our evaluator refuses rather than print one.



Figure 3: One seed, one caption, r swept. Columns: reference, then $r = 0, 0.05, 0.1, 0.3, 0.5, 0.7, 1.0$. The background crowd thins out, the eyewear drifts from the reference frame, and the clothing colour detaches, while identity and pose survive. The reference is never shown to the model.

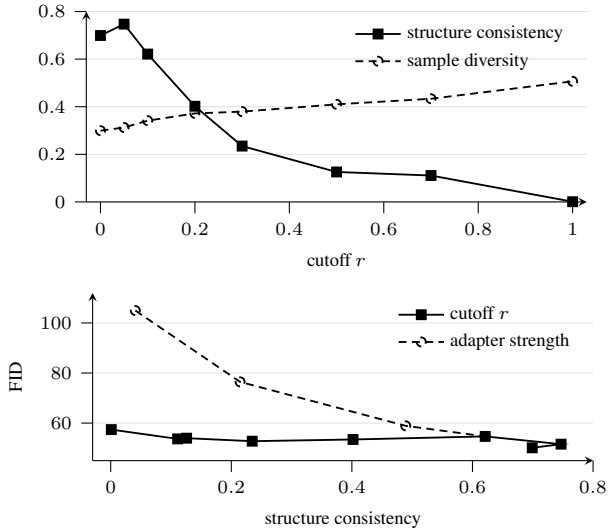


Figure 4: Top: structure consistency falls and sample diversity rises as the cutoff grows. **Bottom:** the same runs in the structure-consistency / FID plane. Narrowing the condition’s bandwidth traverses the axis at nearly constant FID; weakening the adapter buys the same structure consistency by degrading the image. Both panels are plotted directly from the evaluation outputs.

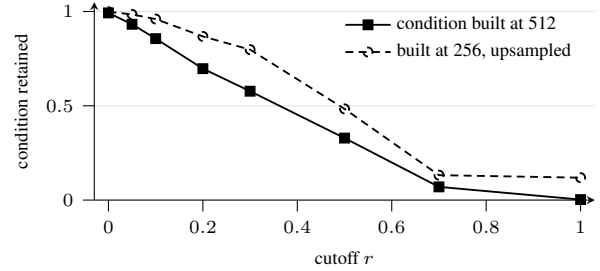


Figure 5: Correlation between the condition fed to the auto-encoder and the same condition re-extracted after an encode-decode round trip, 24 images. Above $r \approx 0.5$ the condition is effectively gone, so an adapter reading the VAE latent cannot recover it. Building the condition at 256 and upsampling puts the structure at lower absolute frequency and preserves far more, which is why comparisons against older pipelines need the source resolution stated.

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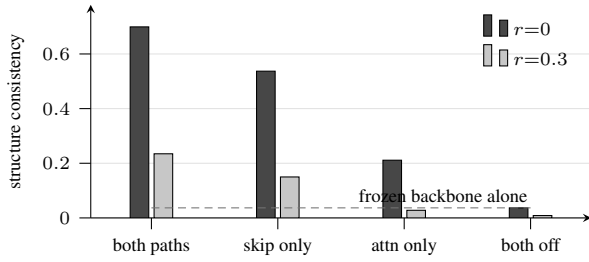


Figure 6: Both injection paths on, then each alone, then neither. The skip residuals carry most of the structure signal; cross-attention alone recovers little of it, and with both off the model falls to the frozen backbone (FID 120.50 against 50.13 with both paths on). The two paths are close to additive.



Figure 7: Hand-drawn face sketches (odd columns) and what the model makes of them (even columns), zero-shot. The drawings carry no cutoff; the model has to read structure density off the drawing itself, which the per-image min-max normalisation of the condition is what permits.

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